Inverse Function $f^{-1}(x)$

Example 1: If $f(x)$ is

x	-2	-1	0	1	2
y	3	2	1	-2	-3

then $f^{-1}(x)$ is

x					
y					

**The domain of $f(x)$ becomes the range of $f^{-1}(x)$. (the x & y values switch)

How to find the inverse $f^{-1}(x)$ *given* $f(x)$. 1. Switch x & y 2. Solve for y

Example 2: Find the inverse of $f(x)$.

A) $f(x)=x+4$ B) $f(x)=\frac{1}{3}x$ C) $f(x)=\frac{5-3x}{2}$

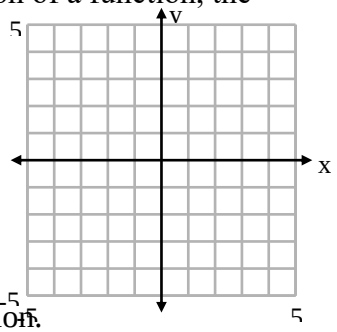
D) $f(x)=\sqrt[3]{x+1}$ E) $f(x)=x^2+1$

Definition of a inverse function: The functions *f* and f^{-1} are inverses of each other if and only if $f\left(f^{-1}(x)\right)=x,$ for every *x* in the domain of *g*, AND $f^{-1}\left(f(x)\right)=x,$ for every *x* in the domain of *f*.

Example 3: For the function $f(x)=\frac{1}{2}x+5$ find:

a) $\left(f\circ f^{-1}\right)(652)$ b) $\left(f^{-1}\circ f\right)\left(-\sqrt{27}\right)$ c) $\left(f^{-1}\circ f\right)(-3.8\pi)$

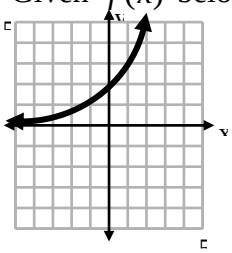
Graph of the inverse function: If we interchange the *x* and *y* 's for every point on the graph of a function, the graph would be reflected over the line **y = x** .

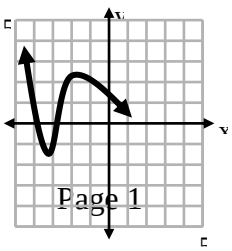


Example 4: Graph $f(x)=x^2$ and its inverse.

Horizontal Line Test: tests the original function, $f(x)$, to see if the inverse is also a function.

Example 5: Given $f(x)$ below, is the inverse a function?

A) 

B) 

C) 